

ME & XE (Hinglish)
Applied Thermodynamics
Analysis of VCRS

DPP: 1

Q 1 The horse power of the motor running the compressor of a refrigerator having COP 4.5 and extracting 200 kJ/kg of evaporating heat with 1.5 kg min^{-1} refrigerant flow rate is

- (A) 0.5 (B) 1
(C) 1.5 (D) 2

Q 2 A reversed Carnot cycle works between temperatures of 270K and 330K. What is the heat rejected by device per minute if work done in compressor is 200 kJ/min?

- (A) 570 kJ (B) 853 kJ
(C) 1100 kJ (D) 912 kJ

Q 3 A reversed Carnot cycle used for chiller plant works between -33°C and 27°C when it is winter. If the plant produces 10 tonne of ice per hour in winter, then what is the amount of ice produced in summer when environment temperature increases by 10 degrees?

- (A) 8.57 tonne (B) 5.64 tonne
(C) 11.00 tonne (D) 9.12 tonne

Q 4 A 2-ton refrigeration unit is used in a refrigerator. If the ratio of minimum to maximum temperature in the cycle is 0.8, then the compressor work if actual COP of cycle is 60% of ideal cycle is _____ kW (round off to two decimal places).

Q 5 A VCR cycle rejects 500kJ of heat per minute. The amount of ice produced at 0°C from water at 30°C if COP of refrigerator is 1.5, is _____ kg/min (round off to two decimal places). Latent heat of fusion water is 334 kJ/kg. Take C of water as 4.2 kJ/kg

Q 6 Two refrigerators A and B operate in series. The refrigerator A absorbs energy at the rate of 1kJ/s from a body at temperature 300 K and rejects energy as heat to a body at temperature T. The refrigerator B absorbs the same quantity of energy which is rejected by the refrigerator A from the body at temperature T, and rejects energy as heat to a body at temperature 1000 K. If both the refrigerators have the same C.O.P., then the temperature T of the body is _____ K (round off to one decimal place).

Q 7 The capacity of a refrigerator is 200 TR when working between -6°C and 25°C . The mass of ice produced per day from water at 25°C is _____ tonnes (round off to two decimal places). Assume that the cycle operates on reversed Carnot cycle and latent heat of ice 335 kJ/kg.

Q 8 A cold storage is to be maintained at -5°C while the surroundings are at 27°C . The heat leakage from the surroundings into the cold storage is estimated to be 29 kW. The actual C.O.P. of the refrigeration plant is one-third of an ideal plant working between the same temperature. The power required to drive the plant is _____ kW (round off to one decimal place).

Q 9 A refrigerant used in vapor compression refrigeration has enthalpy at exit of capillary tube as 135 kJ/kg. If the refrigerant was completely liquid when it enters the expander, then the mass fraction of refrigerant converted to vapour when $h_f = 90 \text{ kJ/kg}$ and latent heat of vaporization is 350 kJ/kg is _____ % (round off to one decimal place).

Q 10 In an idealized Carnot vapor refrigeration cycle, refrigerant is used as the working fluid. The refrigerant enters the condenser as saturated vapor at 28°C and leaves as saturated liquid. The evaporator operates at a temperature of -10°C . The thermodynamic properties of Refrigerant is given in table.

Pressure (kPa)	T_{sat}	Enthalpy (kJ/g)		Entropy (kJ/kg K)	
		h_f	h_g	s_f	s_g
200.74	-10	40	200	0.15	0.9
727.31	28	90	270	0.30	0.8

Which of the following options is/are correct?

- (A) The work input to the compressor is 91.44 kJ/kg.
(B) The work developed by the turbine is 18 kJ/kg.
(C) The work input to the compressor is 80 kJ/kg.
(D) The work developed by the turbine is 14.80 kJ/kg.

Answer Key

Q1	C	Q6	547.7 (545.0 to 550.0)
Q2	C	Q7	13.745 (13.5 to 14.0)
Q3	A	Q8	10.4 (10.0 to 11.0)
Q4	2.92 (2.80 to 3.00)	Q9	12.5 (12.0 to 13.0)
Q5	0.65 (0.60 to 0.70)	Q10	A,B



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ME (Hinglish)

Applied Thermodynamics

Gas Refrigeration

DPP: 1

Q1 A good refrigerant should have which of the following properties.

- (A) High latent heat of vaporization
- (B) Lower specific heat
- (C) High critical temperature
- (D) Low specific volume

Q2 Which of the following is/are true disadvantage(s) of Bell-Coleman cycle

- (A) Low COP
- (B) High running cost
- (C) Small system size
- (D) More chances of frosting at expander

Q3 The chemical formula of R-134 is

- (A) CHClF_2
- (B) $\text{C}_2\text{Cl}_3\text{F}_3$
- (C) $\text{C}_2\text{Cl}_2\text{F}_4$
- (D) $\text{C}_2\text{H}_2\text{F}_4$

Q4 A Bell Coleman cycle works under pressure ratio of 10. The refrigerating effect when air enters compressor at 300K and air is cooled by 40 degrees in condenser is _____ kJ/kg (round off to one decimal place).

Q5 A Bell Coleman cycle works under pressure ratio of 10. Air enters compressor at 300 K and air is cooled by 40 degrees in condenser. The change in COP if air is superheated in the evaporator by 10 degrees _____%.

Q6 Which of the following is true for Refrigerants.

- (A) In Room Air-Conditioner refrigerant is CHClF_2 (mono-chloro - difluoro methane), also called Freon 22 or R22.
- (B) In the Domestic Refrigerator refrigerant is R12, CCl_2F_2 (dichloro - difluoro methane)
- (C) Freons are responsible for the depletion of the ozone layer by chlorine atoms in the upper atmosphere (stratosphere).

(D) All of these

Q7 The chemical name of Freon 22 is

- (A) CCl_3F
- (B) CH_3Cl
- (C) CHClF_2
- (D) CCl_2F_2

Q8 Match List-I (Refrigerant) with List-2 (Uses) and select the correct answer using the code given below the list- II

List – I	List – II
(i) R-22	(A) Direct contact freezing of foods
(ii) R – 134a	(B) Cold storage plants
(iii) NH_3	(C) Windows AC
(iv) Brine	(D) Domestic Refrigerator
(v) CO_2	(E) Milk chilling plants

(A) i-D, ii-C, iii-B, iv-E, v-A

(B) i-C, ii-A, iii-D, iv-E, v-B

(C) i-C, ii-D, iii-B, iv-E, v-A

(D) i-D, ii-C, iii-A, iv-E, v-B

Q9 Which of the following statements is/are true:

- (A) An automatic expansion valve maintains a constant pressure in the condenser.
- (B) An automatic expansion valve maintains a constant pressure in evaporator.
- (C) In an automatic expansion valve the mass flow rate of refrigerant increases as refrigeration load increases.
- (D) Automatic expansion valve-based systems are critically charged.

Q10 The subcooling in a vapour compression refrigeration system can be achieved by which of the following methods.

- (A) Circulating more quantity of cooling water through condenser.



(B) Using water colder than the main circulating water.

(C) Employing a heat exchanger
(D) Any one of the above.



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Answer Key

Q1 (A, B, C, D)

Q2 (A, B)

Q3 (D)

Q4 20.5~21

Q5 0~0

Q6 (D)

Q7 (C)

Q8 (C)

Q9 (B, C)

Q10 (D)



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ME & XE (Hinglish)
Applied Thermodynamics
Psychrometry

DPP: 1

- Q 1** Which of the following will actually tell the amount of water vapour in air?
 (A) Specific humidity
 (B) Relative humidity
 (C) Degree of saturation
 (D) None
- Q 2** What is the relative humidity in the atmospheric air having specific humidity of 16 gm/kg of air and 25°C DBT. P_{vs} at 25°C is 0.03098 bar.
 (A) 82% (B) 78%
 (C) 90% (D) 68%
- Q 3** Air at atmospheric condition of 100kPa and 25°C has 25g of water present per kg of Moist air. The relative humidity of air if saturation pressure at 25°C is 4 kPa is _____% (round off to one decimal place).
- Q 4** A room contains air at 20°C and 98 kPa at a relative humidity of 85 percent. If the saturation pressure at 20°C is 2.3392 kPa and $h = 2537.4$ kJ/kg, what will be the partial pressure of dry air and the enthalpy per unit mass of dry air?
 (A) 96.01 kPa, 52.83 kJ/kg
 (B) 124.8 kPa, 68.68 kJ/kg
 (C) 76.81 kPa, 42.26 kJ/kg
 (D) 53.76 kPa, 32.58 kJ/kg
- Q 5** A moist air sample has dry bulb temperature of 30°C and specific humidity of 12 g water vapour per kg dry air. The degree of saturation (in %) of air sample, if the saturation vapour pressure of water at 30°C is 5 kPa and the total pressure is 90 kPa is _____(round off to one decimal place).
- Q 6** 5 kg of moist air with specific humidity of 0.025 mixes with 10 kg of dry air. The result mixture will have specific humidity of _____gm/kg of d.a. (round off to one decimal place).
- Q 7** There is a room which contains 30 kg of dry air and 0.6 kg of water vapour. The total pressure and temperature of air in the room are 100 kPa and 25°C respectively. ($P_{vs} = 3.17$ KPa; at 25 °C for water) What is the relative humidity of the air in the room (in percentage)?
 (A) 96.25 (B) 97.53
 (C) 98.27 (D) 100
- Q 8** A room is occupied by an air-water vapour mixture at 38°C ($P_s = 0.06624$ bar). The atmosphere pressure is 1 bar and relative humidity is 72%. The humidity ratio is _____ g/kg of dry air.
 (A) 0.03115 (B) 31.15
 (C) 0.3115 (D) 3.115
- Q 9** The atmospheric air at 15° C with density 1 kg/m³ and 1 bar pressure has 80% relative humidity is supplied to the heating chamber at the rate of 100 m³/min. The leaving air has a temperature of 22° C without change in its moisture contents. The heat added to the air is _____kJ/min (round off to nearest integer).
- Q 10** A metallic water container which contains water at room temperature, is cooled slowly by adding ice. When temperature of the container reaches 12.8°C (corresponding vapour pressure 1.479 kN/m²), moisture from the room begins to condense on the container. If barometric pressure of room is 1.01325 bar, then the parts by mass of water vapour (gm w.v/kg mixture) in the room air is _____ (round off to one decimal place).

Answer Key

Q1	A	Q6	8~8.5
Q2	A	Q7	C
Q3	98.5~99.5	Q8	B
Q4	A	Q9	700~725
Q5	33.5~34.5	Q10	8.5~9.5



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ME & XE (Hinglish)
Applied Thermodynamics
Diesel Cycle Analysis

DPP: 1

- Q 1** A four-stroke engine completes one cycle in
 (A) 4 revolutions of crankshaft
 (B) 2 revolutions of crankshaft
 (C) 1 revolution of crankshaft
 (D) 3 revolutions of crankshaft
- Q 2** Choose the correct option(s) from below.
 (A) For the same speed, the number of power strokes in 4-stroke engine is twice that of 2-stroke engine.
 (B) lighter flywheel is used in 2-stroke engines than 4-stroke engine
 (C) In 2-stroke engines, ports are operated by piston movement
 (D) 2-stroke engines are lighter in weight than 4-stroke engines
- Q 3** In an ideal diesel cycle, the clearance volume is 10 percent of stroke and cut-off takes place at 5 percent of stroke. Based on above data which of the following options is/are correct?
 (A) the compression ratio is 10
 (B) the compression ratio is 11
 (C) cut-off ratio is 2.0
 (D) cut-off ratio is 1.5
- Q 4** The maximum pressure and temperature in air standard Diesel cycle are 44 bar and 1600 K respectively. If the minimum pressure and temperature are 1 bar and 300 K then the fuel cut-off ratio is _____(round off to one decimal place).
- Q 5** An engine working on air standard Otto cycle has a cylinder diameter of 10 cm and stroke length of 15 cm. The ratio of specific heats for air is 1.4. If the clearance volume is 196.3 cc and the heat rejected is 1000 kJ/kg, the work output per kg of air is _____kJ (round off to one decimal place).
- Q 6** In an otto engine during compression process, pressure is seen to be 150 kPa at $\frac{1}{8}$ th of stroke and 1.50 MPa at $\frac{7}{8}$ th of stroke. What is the efficiency of this engine?
 (A) 42.5% (B) 54.3%
 (C) 69.8% (D) 48.4%
- Q 7** The compression ratio of the air-standard Otto cycle is 8. The initial conditions of air at the beginning of the compression stroke are 300K and 0.1 MPa. The amount of energy added during the combustion process is 29 kJ/mol. The work done per kg of air is _____kJ (round off to one decimal place).
- Q 8** The percentage loss in the ideal efficiency of a diesel engine with compression ratio 14 if the fuel cut-off is delayed from 5% to 8%, is _____% (round off to two decimal places).
- Q 9** In a diesel engine with variable compression ratio, the initial compression ratio is 16 : 1. The ratio of specific heats is 1.4. For the same cut-off ratio of 4.0, if the compression ratio is increased by 25 %, the increase in air standard thermal efficiency of the engine will be _____% (round off to two decimal places).
- Q 10** A two-stroke square ratio engine runs with speed of 3000 rpm and fuel consumption of 5 litres/hr. The fuel has specific gravity of 0.7 and air-fuel ratio is 19. The mean piston speed is 600 m/min. The ambient conditions are 1.013 bar, 15°C. What is the volumetric efficiency considering R for gas = 0.287 kJ/kg-K?
 (A) 38.4% (B) 26.4%
 (C) 51.6% (D) 42.0%

Answer Key

Q1 B
Q2 B,C,D
Q3 B,D
Q4 1.5~2
Q5 1150~1190

Q6 C
Q7 550~600
Q8 3~3.5
Q9 7.5~8
Q10 A



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Mechanical Engineering (Hinglish)

Applied Thermodynamics

DPP: 1

Engine Tests and Analysis

- Q 1** What is the octane number of a fuel if a fuel mixture contains 30gm of iso-octane and 5 gm of n-Heptane.
 (A) 56.52 (B) 85.71
 (C) 80.42 (D) 75.00
- Q 2** In a Morse test on 3 cylinders, 2- stroke engine SI engine, the brake power is 10 kW and the BHP of individual cylinder with spark cut-off are 6.0KW, 5.9KW, and 5.95KW respectively. The mechanical efficiency of the engine is _____% (round off to one decimal place).
- Q 3** A rope dynamometer is used to measure power of an IC engine. If the rope is wrapped by 2 turns on the dynamometer shaft, then the power developed by engine when tension in the slack and tight side of rope is 50 and 500N, is _____kW (round off to one decimal place). The drum radius is 0.5 m and engine rotate at 3000 RPM
- Q 4** A motoring test was performed on engine to calculate frictional power of engine. If the motor consumes 10A of current at potential difference of 220V what is the frictional power? Assume motor has 80% efficiency.
 (A) 1.76 kW (B) 2.20 kW
 (C) 1.53 kW (D) 2.00 kW
- Q 5** A brake dynamometer is used to measure power developed by engine. It measures a torque of 500N-m at engine speed of 900 RPM. The torque measured by this dynamo if engine speed is dropped to 600 RPM on reducing the fuel supply by 25%, is _____N-m (round off to nearest integer). Frictional power of engine is 2kW.
- Q 6** 44 kg of C_3H_8 is burnt with air (Mol. Wt. = 29) to produce 44 kg of CO_2 and remaining CO. What is the approximate amount of air used in the combustion?
 (A) 56 kg (B) 220 kg
 (C) 423 kg (D) 550 kg
- Q 7** A fuel containing 47% C, 42% Hydrogen, 10% oxygen and 2% Sulphur by weight is burnt in presence of Air. If the mass of fuel is 1 kg then the mass of air required so that Sulphur produces SO_2 , is _____kg (round off to one decimal place).
- Q 8** 1kg of fuel consisting of a mixture of 50% methane and 50% hydrogen by volume is burnt in presence of oxygen. The amount of oxygen required for complete combustion is _____kg (round off to two decimal places).
- Q 9** The Willan's line measured for a four-stroke, four-cylinder is expressed as: $FC = 0.4 + 0.1 \cdot BP$, Where FC is the rate of fuel consumption in gm/s and BP is the brake power in kW. If the engine consumes 1 gm/s of fuel, then the indicated thermal efficiency is _____% (round off to nearest integer). Take CV of fuel = 40MJ/kg.
- Q 10** During trial of four stroke single cylinder engine the load on dynamometer is found 20 kg at radius of 50 cm. The speed of rotation is 3000 rpm. The bore and stroke are 20 cm and 30cm respectively. Fuel is supplied at the rate of 0.15 kg/min. The calorific value of fuel may be taken as 43 MJ/kg. After some time, the fuel supply is cut and the engine is rotated with motor which required 5 kW to maintain the same speed of rotation of engine. Based on above data, which of the following options is/are correct?
 (A) The mechanical efficiency is 86%
 (B) The mechanical efficiency is 77%
 (C) The bsfc is 0.15
 (D) The bsfc is 0.29

Answer Key

Q1 B
Q2 82~82.5
Q3 70~71
Q4 A
Q5 550~560

Q6 D
Q7 19~20
Q8 4.2~4.6
Q9 24~26
Q10 A,D



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Mechanical Engineering (Hinglish)

Applied Thermodynamics

DPP: 1

Brayton Cycle

- Q 1** Which of the following is/are correct with respect to gas turbines?
- (A) It usually operates as an open cycle.
 (B) The working fluid undergo no phase change.
 (C) It can work with low quality fuel.
 (D) It is used in aircrafts due to its light weight.
- Q 2** A gas turbine plant operates on Brayton cycle between $T_{\min} = 300 \text{ K}$ and $T_{\max} = 1000^\circ\text{C}$. Optimum Pressure ratio of the cycle will be _____ (round off to one decimal place).
 [Take γ for gas = 1.4]
- Q 3** A gas turbine plant operates on Brayton cycle between 300 K and 1300 K. Select the correct option(s) from below.
- (A) The maximum cycle efficiency is 76.9 %
 (B) The maximum cycle efficiency is 53.2%
 (C) Maximum work done per kg of air is 352.75 kJ/kg
 (D) Maximum work done per kg of air is 167.26 kJ/kg
- Q 4** Atmospheric air at 101.325 kPa and 27°C enters the compressor of a gas turbine operating on Brayton cycle which is having a pressure ratio of 6 ($\gamma_{\text{air}} = 1.4$). If the turbine work is 2.5 times the work of compressor, then the maximum temperature in the cycle will be _____ K (round off to nearest integer).
- Q 5** An ideal Brayton cycle, operating between the pressure limits of 1 bar and 8 bar, has minimum and maximum temperatures of 300 K and 1500 K. The difference between the temperatures at the end of compression and expansion processes is _____ K (round off to the nearest integer).
- Q 6** A gas plant operates on Brayton cycle between temperature limits of 300 K and 1000 K. The efficiency of the cycle corresponding to optimum pressure ratio is (take $\gamma = 1.4$).
- (A) 73% (B) 58%
 (C) 45% (D) 62%
- Q 7** In an ideal Brayton cycle, atmospheric air (ratio of specific heats, $c_p/c_v = 1.4$, specific heat at constant pressure = 1.005 kJ/kgK) at 1 bar and 300 K is compressed to 8 bar. The maximum temperature in the cycle is limited to 1500 K. If the heat supplied at the rate of 100 MW, the mass flow rate (in kg/s) of air required in the cycle is _____ (round off to nearest integer).
- Q 8** A simple Brayton cycle has a pressure ratio of 6 and a maximum temperature of 900 K. Air enters the compressor at 100 kPa, 300 K. Based upon cold-air standard analysis assumptions, the work ratio of this cycle is?
- (A) 0.44 (B) 0.56
 (C) 0.65 (D) 0.72
- Q 9** An ideal gas turbine working on Brayton cycle has minimum and maximum temperatures 310 K and 1300 K respectively and back work ratio is 0.50. What will be the temperature at the end of compression.
- (A) 735 K (B) 650 K
 (C) 545 K (D) 462 K
- Q 10** In a gas turbine cycle the temperature at end of compression is 500 K and peak temperature in cycle is 1400 K. The mean temperature at which heat is supplied is
- (A) 950 K (B) 786 K
 (C) 874 K (D) 702 K

Answer Key

Q1	A,B,D	Q6	C
Q2	12~13	Q7	100~110
Q3	A,C	Q8	A
Q4	1240~1270	Q9	B
Q5	275~295	Q10	C



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Mechanical Engineering (Hinglish)

Applied Thermodynamics

DPP: 1

Rankine Cycle

- Q1** Rankine cycle is different from Carnot cycle during.
 (A) Expansion process
 (B) Heat rejection process
 (C) Heat addition process
 (D) Pump work
- Q2** Superheating of steam before expansion in a Rankine cycle
 (A) Decreases thermal efficiency
 (B) Has no effect on thermal efficiency
 (C) Increases thermal efficiency
 (D) May increase or decrease thermal efficiency
- Q3** Rankine efficiency of a steam power plant
 (A) Improves in summer as compared to that in winter
 (B) Improves in winter as compared to that in summer
 (C) Is unaffected by climatic conditions
 (D) None of these
- Q4** In a Rankine cycle which of the following effects are not true when condenser pressure is decreased?
 (A) Efficiency increases
 (B) Pump work increases
 (C) Work output increases
 (D) Life of turbine blades increases
- Q5** A Gas turbine working on Brayton cycle has pressure ratio of 4. Inlet condition of compressor is 1 Bar and 27 °C. If there is a heat loss of 10 kJ/kg from the compressor, the work required to run compressor is ____ kJ/kg (round off to nearest integer). Assume $c_p = 1 \text{ kJ/kgK}$ & $\gamma = 1.4$
- Q6** A Gas turbine working on Brayton cycle has pressure ratio of 4. Combustion products enter turbine at 1500 K. If there is a heat loss of 20 kJ/kg from the Turbine, then the work developed by turbine ____ kJ/kg (round off to nearest integer). Take $c_p = 1.1 \text{ kJ/kgK}$ & $\gamma = 1.33$.
- Q7** A Gas turbine working on Brayton cycle has pressure ratio of 4. Air enters compressor at 300 K. If air fuel ratio is 80 and combustion efficiency is 96%, the temperature of air entering turbine if calorific value of fuel is 41 MJ/kg, is ____ K (round off to nearest integer). Take $c_p = 1 \text{ kJ/kgK}$ & $\gamma = 1.4$
- Q8** The steam consumption of steam engine is 20 tonnes per shift of 8 hours when developing 370 kW. The specific enthalpies at the outlet and inlet of the boiler are 2778.0 kJ/kg and 191.8 kJ/kg. The thermal efficiency of engine is ____% (round off to one decimal place).
- Q9** A power plant working on ideal Rankine cycle operates between pressure limits of 180 bar and 0.06 bar. At condenser pressure, the specific enthalpy and specific volume of saturated liquid are 151.5 kJ/kg and $0.0010064 \text{ m}^3/\text{kg}$ respectively. The pump work is ____ kJ/kg (round off to one decimal place).
- Q10** The power developed by a turbine in ideal Rankine cycle is 1200 kW. The heat supplied to the boiler and the heat rejected in condenser are 3400 kJ/kg and 2800 kJ/kg respectively. If the energy required by the feed pump is 36 kW, then the required mass flow rate of steam is ____ kg/s (round off to two decimal places).



Answer Key

Q1 (C)
Q2 (C)
Q3 (B)
Q4 (D)
Q5 150~160

Q6 455~465
Q7 930~945
Q8 20~21
Q9 17.8~18.4
Q10 1.9~2



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Mechanical Engineering (Hinglish)

Applied Thermodynamics

DPP: 2

Rankine Cycle

Q1 Which of the following statements regarding effects of reheating of steam in a steam turbine is true:

- (A) It decreases the specific output of the turbine.
- (B) It decreases cycle efficiency.
- (C) It increases blade erosion.
- (D) It improves the quality of exit steam.

Q2 In the Rankine cycle, with the maximum steam temperature being fixed from metallurgical consideration, as the boiler pressure increases,

- (A) The condenser load will increase
- (B) The quality of turbine exhaust will decrease
- (C) The quality of turbine exhaust will increase
- (D) The quality of turbine exhaust will remain Unchanged

Q3 A steam powerplant works with boiler pressure of 180 bar and condenser pressure of 0.1bar. The enthalpy at inlet to boiler is 223 kJ/kg and at exit of boiler is 3200 kJ/kg. Reheat is employed at pressure of 45 bar with enthalpy of cold reheat as 2900 kJ/kg and that of hot reheat as 3445 kJ/kg. Enthalpy at exit of low pressure turbine is 2500 kJ/kg. Neglecting pump work, the efficiency of cycle is _____% (round off to one decimal place).

Q4 Steam powerplant generates 50MW of power while running at efficiency of 35%. If the coal used in powerplant has a calorific value of 28 MJ/kg and boiler efficiency is 80%, then the amount of coal required per day to meet power requirements of the plant is _____tonnes/day (round off to nearest integer).

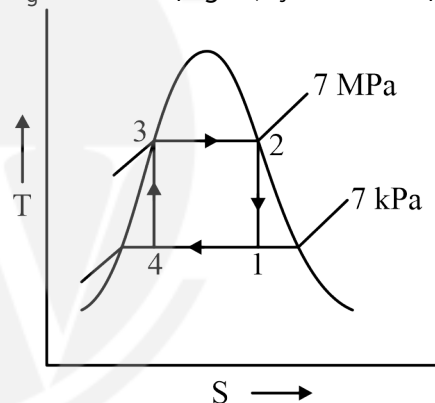
Q5 The steam consumption of steam engine is 20 tonnes per shift of 8 hours when developing

370 kW. The specific enthalpies at outlet and inlet of the boiler are 2778.0 kJ/kg and 191.8 kJ/kg. The thermal efficiency of engine is _____% (round off to one decimal place).

Q6 A Carnot cycle works on steam between the pressure limits of 7 MPa and 7 kPa. The thermal efficiency of the cycle is _____% (round off to one decimal place).

At 7 MPa: $h_f = 1267$ kJ/kg, $h_g = 2772.1$ kJ/kg, $s_g = 5.8133$ kJ/kg · K, $s_f = 3.1211$ kJ/kg · K

At 7 kPa: $h_f = 162.60$ kJ/kg, $h_g = 2572.14$ kJ/kg, $s_g = 8.2801$ kJ/kg · K, $s_f = 0.5564$ kJ/kg · K



Q7 In a steam power plant working on ideal Rankine cycle, the steam turbine receives steam at 10 bar 2500°C and discharge at 0.5 bar. The enthalpies at inlet and outlet of the turbine are 2942 kJ/kg and 2408 kJ/kg respectively. If enthalpy at pump entry is 340 kJ/kg, neglecting pump work the efficiency of the power plant is _____% (round off to one decimal place).

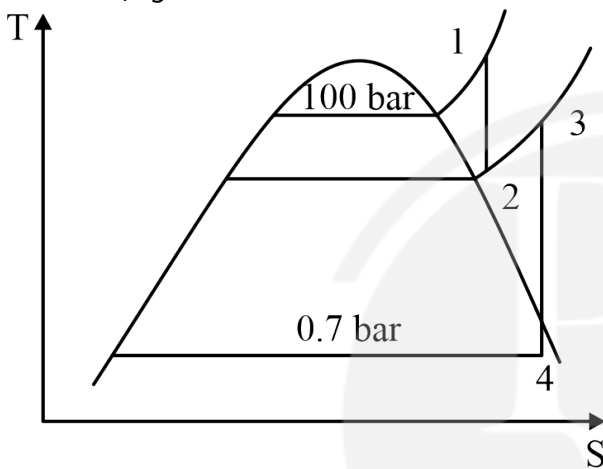
Q8 A power plant working on a Rankine cycle works between pressure limits of 12 bar and 0.7 bar. The enthalpy and entropy at turbine inlet are 3432 kJ/kg and 7.0198 kJ/kg · K.

At 0.7 bar properties of steam are $s_f = 0.5589$ kJ/kg · K, $s_g = 7.459$ kJ/kg · K, $h_f = 163$ kJ/kg, h_g


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=2246 kJ/kg. The enthalpy of the wet steam at turbine exit is ____ kJ/kg (round off to nearest integer).

- Q9** Steam at 100 bar and 500°C ($h_1 = 3373.6$ kJ/kg) enters a steam turbine that has one stage of reheat. The steam is exhausted from the turbine at 0.7 bar and 85% dry. The net work developed by the turbine is 1600 kJ/kg of steam. Neglecting pump work, the heat supplied in reheater ____ kJ/kg (round off to nearest integer). At 0.7 bar, $h_f = 163.38$ kJ/kg & $h_{fg} = 2409.2$ kJ/kg.



- Q10** In a Reheat Rankine Cycle, the enthalpies at various states are given below.

$$h_1 = 3336.5 \text{ kJ/kg}$$

$$h_2 = 2814.4 \text{ kJ/kg}$$

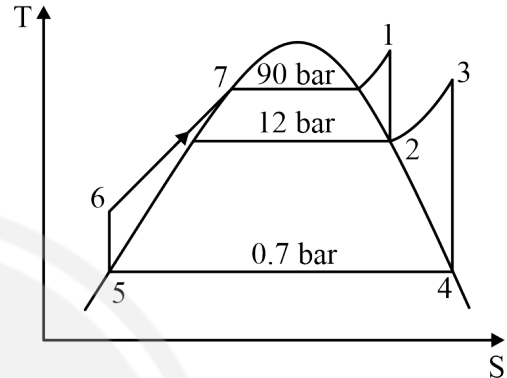
$$h_3 = 3432.8 \text{ kJ/kg}$$

$$h_4 = 2367.7 \text{ kJ/kg}$$

$$h_5 = 163.4 \text{ kJ/kg}$$

$$h_6 = 172.4 \text{ kJ/kg}$$

The efficiency of the cycle is ____ % (round off to one decimal place).



Answer Key

Q1 (D)
Q2 (B)
Q3 35~36
Q4 545~555
Q5 20~21

Q6 43.8~45
Q7 20~21
Q8 2000~2200
Q9 435~440
Q10 41~42



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ME & XE (Hinglish)
Applied Thermodynamics
Centrifugal Air Compressors

DPP: 1

- Q 1** Centrifugal compressors are used where
 (A) high pressure and low volume are required
 (B) low pressure and high-volume flow are required
 (C) high pressure and high-volume flow are required
 (D) low pressure and low-volume flow are required
- Q 2** A centrifugal air compressor with radial blades runs at 15000 RPM. If the inlet diameter is 15cm and exit diameter is 30cm. If inlet temp is 17 °C, then the temperature produced at exit of compressor is _____ °C (round off to two decimal places).
- Q 3** A centrifugal air compressor having a pressure ratio of 5, compresses air at a rate of 10 kg/s. If the initial pressure and temperature are 1 bar and 20°C respectively, then what is the power required to drive the compressor? Take $\gamma = 1.4$ and $c_p = 1 \text{ kJ/kg.K}$.
 (A) 1550 kW (B) 1224 kW
 (C) 1710 kW (D) 934 kW
- Q 4** In a centrifugal compressor running at 15000 RPM, diameter and whirl velocity at outlet are 500 mm and 380 m/s. The degree of reaction of the turbine is.
 (A) 51.6% (B) 45.3%
 (C) 55.8% (D) 61.5%
- Q 5** A centrifugal air compressor runs at 15000 RPM. The inlet diameter is 15cm and exit diameter is 30cm. The slip factor is 0.9 and work input factor is 1.05. If 5 kg/min of air passes through compressor, then the power required to run compressor is _____kW (round off to two decimal places).
- Q 6** A centrifugal air compressor runs at 27000 RPM. The blade diameter at exit is 20 cm and isentropic efficiency of compressor is 90%. If air enters compressor at 22 °C, then the pressure ratio developed in the compressor is _____ (round off to two decimal places).
- Q 7** A centrifugal air compressor runs at 20000 RPM. Air enters at 300K & 1 bar. Diameter of inlet of compressor is 10cm and that of exit of blade is 20 cm and there is pre-whirl of 15 degrees provided at inlet. The work required per kg of air to run the compressor is _____kJ/kg (round off to one decimal place).
- Q 8** A centrifugal air compressor runs at 12000 RPM. Air enters at 300K & 1 bar. Diameter at the inlet of blade is 10cm and that at exit is 30cm. Blade angle at exit is 70 degrees. If velocity of flow is constant at 50m/s, then the work required per kg of air to run the compressor is _____kJ/kg (round off to one decimal place).
- Q 9** A centrifugal compressor runs at a speed of 15000 rpm and delivers 30 kg of air per second. The work done on the compressor is 28.8 kJ/kg of air. The torque on the compressor is _____ kN-m (round off to two decimal places).
- Q 10** A centrifugal compressor handles 540 kg/min of air at a speed of 20,000 rpm. The air enters the impeller eye with no pre-whirl. The ratio of whirl speed to tip speed is 0.9. The diameter of tip is 426 mm. then the power required is _____kW (round off to nearest integer).

Answer Key

Q1 B
Q2 70~75
Q3 C
Q4 A
Q5 4.2~4.6

Q6 2~2.3
Q7 40~42
Q8 31.8~32.4
Q9 0.5~0.6
Q10 1575~1650



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ME & XE (Hinglish)
Applied Thermodynamics
Axial Compressor

DPP: 1

- Q 1** If an axial flow compressor is designed for a constant velocity through all stages, the area of the annulus of the succeeding stages will
 (A) Remain the same
 (B) Progressively decrease
 (C) Progressively increase
 (D) Depend upon the number of stages
- Q 2** While flowing through the rotor blades in an axial flow air compressor, the relative velocity of air
 (A) continuously decreases
 (B) continuously increases
 (C) first increases and then decreases
 (D) first decreases and then increases
- Q 3** In an axial flow compressor stage, air enters and leaves the stage axially. If the axial component of the air leaving the rotor is half the mean peripheral velocity of the rotor blades and air leaves rotor blade axially, then the degree of reaction will be
 (A) 1.00 (B) 0.75
 (C) 0.50 (D) 0.25
- Q 4** In an axial flow air compressor: $u = 200$ m/s, $v_f = 180$ m/s, $\alpha_1 = 50^\circ$, $\alpha_2 = 15^\circ$ and $\rho = 1$ kg/m³. The work done per stage per kg of air is ____ kJ (round off to one decimal place).
- Q 5** In an axial compressor, the mean blade speed is 220 m/s and axial velocity is 180 m/s. The blade inlet and outlet angles are 42° and 20° respectively. The degree of reaction of the compressor is ____% (round off to one decimal place).
- Q 6** In an axial flow compressor with 50% degree of reaction, the relative air angles at rotor inlet and outlet are 130° & 100° respectively. Blade velocity = 180 m/s. The work done per stage is ____ kJ/kg (round off to one decimal place).
- Q 7** An axial compressor is fitted with half-reaction blading, the blade inlet and outlet angles being 50° and 15° when measured from axial direction. The mean diameter of a certain blade pair is 85 cm and the rotor run at 5500 rpm. The whirl speed at inlet is ____ m/s (round off to one decimal place).
- Q 8** A 10 stage axial compressor provide overall pressure ratio of 10 at overall isentropic efficiency of 90%. Air enters at 300K and work is divided equally between all stages. If blade outlet angle is 20 degree, blade mean velocity is 200m/s and flow velocity is 100m/s, then the blade inlet angle is ____ degree (round off to one decimal place).
- Q 9** An axial compressor runs at 15000 RPM. Mean diameter of blade is 0.4m. If the flow velocity is constant to 50m/s, and air is deflected from absolute angle of 30 degree to 60 degree in the flow direction, the work required to run compressor is ____ kJ/kg (round off to one decimal place).
- Q 10** In an axial flow compressor, the axial velocity is uniform across the stage and equal to 120 m/s and mean blade speed of each stage is 200 m/s. If the compressor has symmetrical blades and work done per stage is 22.39 kJ/kg then the blade inlet and exit angles are
 (A) $\beta_1 = 42.30^\circ$, $\beta_2 = 37.13^\circ$
 (B) $\beta_1 = 42.30^\circ$, $\beta_2 = 15.50^\circ$
 (C) $\beta_1 = 52.43^\circ$, $\beta_2 = 20.15^\circ$
 (D) $\beta_1 = 54.33^\circ$, $\beta_2 = 15.30^\circ$

Answer Key

Q1 B
Q2 A
Q3 D
Q4 33~33.6
Q5 51~52.5

Q6 20.7~21.6
Q7 44~46
Q8 62~63
Q9 17~19
Q10 C



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ME & XE (Hinglish)

Applied Thermodynamics

Steam Turbine

DPP: 1

- Q 1** Which of the following statement is correct for steam turbine?
- (A) It converts pressure energy into electrical energy
(B) It converts thermal energy into mechanical energy
(C) It converts thermal energy into kinetic energy
(D) It converts low grade energy into high grade energy
- Q 2** The main aim of compounding steam turbine is to
- (A) improve efficiency
(B) reduce steam consumption
(C) reduce motor speed
(D) reduce turbine size
- Q 3** Which of the following is not the correct statement for steam turbines
- (A) In impulse turbine the cross-section between blades is constant.
(B) Blade manufacturing of reaction turbines is complex.
(C) In reaction turbines the pressure drop takes place only in moving blades.
(D) Impulse turbines are used for small power generation.
- Q 4** Steam expands from enthalpy of 3400kJ/kg to 3200kJ/kg in a nozzle. The velocity of steam coming out of the nozzle will be?
- (A) 200 m/s (B) 632 m/s (C) 836 m/s (D) 14.14 m/s
- Q 5** A combined cycle powerplant works with Brayton cycle as top cycle and Rankine cycle as bottom cycle. If efficiency of both the cycles is 40% then efficiency of combined cycle will be _____% (round off to nearest integer).
- Q 6** In a simple impulse turbine, the nozzle angle at the entrance is 30° . What will be the blade-speed ratio for maximum diagram efficiency?
- (A) 0.43 (B) 0.25
(C) 0.5 (D) 0.75
- Q 7** A D'Laval turbine has absolute velocity at entry and exit to blade as 250m/s and 150 m/s. The work developed by turbine per kg of steam flow will be _____kJ/kg (round off to nearest integer).
- Q 8** A four-row velocity compounded steam turbine develops 6400 kW. What is the power developed by the last row?
- (A) 200 kW (B) 400 kW
(C) 800 kW (D) 1600 kW
- Q 9** A parson's turbine has absolute and relative velocity at entry to blade as 250m/s and 150 m/s with nozzle angle as 30 degree. The work developed by turbine will be _____kJ/kg (round off to nearest integer).

Answer Key

Q1 B,D
Q2 C
Q3 C
Q4 B
Q5 62~66

Q6 A
Q7 19~21
Q8 B
Q9 39~41



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ME & XE(Hinglish)
Applied Thermodynamics
Compressible Flow

DPP: 1

- Q 1** Space shuttle while entering the atmosphere from space stations travels at Mach Number of 5. If the atmospheric temperature at a given location is -50°C , temperature of air at nose of the shuttle will be _____ $^{\circ}\text{C}$ (Round off to nearest integer).
- Q 2** Which of the following statements regarding convergent divergent nozzles are correct.
 (A) Supersonic velocities can be achieved in these nozzles
 (B) They have convergent subsonic and divergent supersonic sections
 (C) Shockwaves may be formed in divergent section
 (D) They are fitted where pressure ratios are low
- Q 3** An object travels in air of pressure 1 bar (abs) at 10°C . The speed of the projectile is 1500 km/hour. The Mach angle is _____ degree (round off to nearest integer).
- Q 4** Which of the following statements are correct for shock waves.
 (A) Pressure increases more across a normal shock wave as against weak shockwave
 (B) Total temperature remains same across shockwaves
 (C) Strong shocks are formed at high Mach numbers
 (D) Shock waves are always produced at supersonic speeds
- Q 5** In a nozzle designed for maximum discharge conditions, the flow velocity in the convergent section of the nozzle is
 (A) Subsonic (B) Sonic
 (C) Supersonic (D) Depends on initial pressure and other conditions of the gas
- Q 6** In a normal shock in a gas
 (A) the stagnation pressure remains the same on both sides of the shock
 (B) the stagnation density remains the same on both sides of the shock
 (C) the stagnation temperature remains the same on both sides of the shock
 (D) the Mach number remains the same on both sides of the shock
- Q 7** Air is expanded in nozzle of gas turbine from pressure of 15 bar to 3 bar. If the process is isentropic, Mach number at exit of nozzle will be _____ (round off to two decimal places).
- Q 8** An airplane is flying at a height of 15 km where the temperature is -50°C . The speed of the plane corresponds to $M = 2.0$. Assuming $k = 1.4$ and $R = 287 \text{ J/kg-K}$, the speed of the plane is _____ km/hr (Round off to nearest integer).
- Q 9** An airplane is moving in an atmosphere with pressure $p = 44 \text{ kPa (abs)}$ and density $\rho = 0.63 \text{ kg/m}^3$. A pilot tube on the plane records the stagnation pressure as 70 kPa (abs). The speed of the airplane is _____ m/s (Round off to nearest integer). ($k = 1.4$ and $R = 287 \text{ J/kg-K}$)
- Q 10** Air at 8.6 bar, and 190°C expands at a rate of 4.5 kg/s through a convergent–divergent nozzle to an atmospheric pressure of 1.013 bar. The critical pressure is _____ 4.54 (4.30 to 4.70) _____ bar (Round off to two decimal places).

Answer Key

Q1	1060~1070	Q6	C
Q2	A,B,C	Q7	1.6~1.8
Q3	53~55	Q8	2150~2160
Q4	A,B,D	Q9	260~266
Q5	A	Q10	4.3~4.7



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